SIW Filter — Plan

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Plan — from here to a fabricated board

Current state: `STATE.md`. Superseded plans: `docs/archive/`.

Two things stand between the design and a board worth ordering:

1. **Bandwidth is ~2× spec** (about 300 MHz vs 143.5 MHz) — a cavity-coupling problem, unaffected by any fabrication choice.
2. **Nothing has been optimised under the CPWG launch**, which is now what dual-mode actually simulates.

Step 1 addresses both at once.

Step 1 — Re-optimise under the CPWG launch ← *next*

Two stages, because PEC solves are cheaper than full-stack ones and there is no reason to pay for the fabrication mesh while the search is still coarse.

1a. Global search, PEC metal

```
siw design --topology dualmode --N 2 --evals 45 --patience 14 --np 8
```

All five geometry parameters move (`a, w_iris, x_iris, y_wall, w_ext`). Bandwidth is already a two-sided term in `score()` (`w_bw=3.0`), so it is targeted, not merely reported.

1b. Polish, full fabrication stack

```
siw design --topology dualmode --N 2 --seed <1a best> --span 2 \
           --patience 14 --evals 30 --np 8 --enig --real-metal
```

`--enig` applies real copper + ENIG + soldermask + silkscreen + open vias to every evaluation. `fixed_extras()` keeps these out of the search box — they are conditions, not dimensions — so 1a and 1b stay comparable.

Done when: return loss at most −16 dB **and** bandwidth within about 10 % of 143.5 MHz, both under the full stack, with 2 reflection zeros.

If bandwidth stalls (it has resisted several searches before): stop searching and calibrate. Measure `k12(w_iris)` and `Qe(w_ext)` on small structures, invert for the 5 % FBW targets (`k12 about 0.069`, `Qe about 16.9`), and assemble. This is the method that worked for the cascade after black-box search failed; the failure mode is that bandwidth is *discontinuous* in the external coupling, which a smooth-function optimiser cannot model.

Check every result before believing it. Insertion loss can never beat the PEC reference. Pole count must be 2. See `STATE.md`, "the one habit that matters".

Step 2 — Lock the fabrication spec

Mostly decided already; confirm the one open item.

Choice	Decision	Basis
Finish	**ENIG**	board is unsoldered and exposed; gold/nickel is a real corrosion barrier, OSP is a pre-solder film
Vias	**open, unfilled**	measured < 0.03 dB difference; do not pay for filling
Silkscreen	**free choice**	electrically irrelevant
Soldermask over the feed	**open it — to be confirmed**	reasoned from mechanism; the confirming run was cancelled and never completed

Run the remaining confirmation:

```
siw sweep --topology dualmode --N 2 --params <final geometry> --real-metal --enig
# versus the same geometry with the mask opening over the feed
```

Step 3 — Produce fabrication outputs

- CST `.bas` export (`cst_export.py`) and a visual check in CST.
- Gerbers with the soldermask opening per Step 2.
- JLCPCB DRC re-check against `config.py`'s recorded capabilities.
- Save the final design as a golden entry tagged `cpwg_taoglas`.

Step 4 — Order the board

Only once Step 1's acceptance criteria are met, or the deviation is explicitly accepted and written down. Do not ship a board that silently misses spec.

Later

- **Paper** — "AI-Driven Design for SIW Filters". The strongest material is already collected: black-box optimisation failing on a *structural* obstacle (a discontinuous objective) rather than a tuning one; the seven-bug taxonomy of silent modelling failures; and honest negative results (the N=4 compact cavity, the via-fill null with its stated mechanism).
- **Mesh convergence study** for `fc` — never done; would put an error bar on every frequency reported.

Do not redo

- **Compact N=4 dual-mode cascade** — five structurally different mechanisms plus a joint search all hit the same ceiling. Raw data in `docs/geometry_exploration/`.
- **Lossy-conductor boundary conditions on internal surfaces** — Palace's `Conductivity` BC is exterior-only. Use a volumetric conductor domain instead; the boundary-tag approach silently produces invalid results.
- **A second launch definition.** There is one, in `topologies/launch.py`. Forking it caused three separate bugs. If a genuinely different launch is ever needed, add a new named dict and tag its results distinctly.